$$C\,(s) + O^{2-}\,(melt) \longrightarrow CO + 2e^-$$

$$C\,(s) + 2O^{2-}\,(melt) \longrightarrow CO_2 + 4e^-$$

Due to the above two reactions, anodes are slowly consumed during the electrolysis. The pure aluminium is formed at the cathode and settles at the bottom. The net electrolysis reaction can be written as follows.

$$4Al^{3+}\,(melt) + 6O^{2-}\,(melt) + 3C\,(s) \longrightarrow 4Al\,(l) + 3CO_2\,(g)$$

Evaluate yourself 4

4. Metallic sodium is extracted by the electrolysis of brine (aq. NaCl). After electrolysis the electrolytic solution becomes basic in nature. Write the possible electrode reactions.

1.6 Refining process

Generally the metal extracted from its ore contains some impurities such as unreacted oxide ore, other metals, nonmetals etc...Removal of such impurities associated with the isolated crude metal is called refining process. In this section, let us discuss some of the common refining methods.

1.6.1 Distillation

This method is employed for low boiling volatile metals like zinc (boiling point 1180 K) and mercury (630 K). In this method, the impure metal is heated to evaporate and the vapours are condensed to get pure metal.

1.6.2 Liquation

This method, is employed to remove the impurities with high melting points from metals having relatively low melting points such as tin (Sn; mp= 904 K), lead (Pb; mp=600 K), mercury (Hg; mp=234 K), and bismuth (Bi; mp=545 K). In this process, the crude metal is heated to form fusible liquid and allowed to flow on a sloping surface. The impure metal is placed on sloping hearth of a reverberatory furnace and it is heated just above the melting point of the metal in the absence of air, the molten pure metal flows down and the impurities are left behind. The molten metal is collected and solidified.

1.6.3 Electrolytic refining:

The crude metal is refined by electrolysis. It is carried out in an electrolytic cell containing aqueous solution of the salts of the metal of interest. The rods of impure metal are used as anode and thin strips of pure metal are used as cathode. The metal of interest dissolves from the anode, pass into the solution while the same amount of metal ions from the solution will be deposited at the cathode. During electrolysis, the less electropositive impurities in the anode, settle down at the bottom and are removed as anode mud.

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